

Parthan Olikkal

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SUMMARY

Researcher specializing in Reinforcement learning, Multimodal AI, and Human-AI Interaction. Develops adaptive decision-making systems using representation learning techniques across vision, wearable sensing, and robotic platforms. Experienced in PyTorch, deep learning, and translating research innovations into scalable intelligent systems.

EDUCATION

University of Maryland Baltimore County

Aug 2021 – Present

Ph.D. Candidate in Computer Science (Dissertation: Synergy-based learning in Human Robot Interaction)

University of Maryland Baltimore County

May 2019 – Aug 2021

M.S. in Computer Science (Thesis: Kinematic and Muscle Synergies in Grasping Hand)

PROGRAMMING SKILLS

- **Languages & Tools:** Python, C++, MATLAB, Simulink, SQL, NoSQL, AWS (Practitioner), Git, Perforce
- **Libraries & Frameworks:** PyTorch, Scikit-learn, MediaPipe, OpenCV, Gym, Isaac Lab, MuJoCo, Unity, Stable-Baselines3
- **ML, DL & RL:** Classical ML models, Autoencoders, Transformers, CNNs, Decomposition Algorithms (PCA, NMF, RPCA, ICA), PPO, DQN, Q-learning, SAC, multi-agent RL

SELECTED RESEARCH EXPERIENCE

UMBC Sensorimotor Control Lab | Researcher (RL & Adaptive AI)

Aug 2021 – Present

Project 1: Adaptive Representation Selection using Proximal Policy Optimization

- Improved reconstruction accuracy by 18% over fixed representation baselines by training a PPO agent to **dynamically select optimal latent feature** subsets from multimodal wearable+vision streams across 10 subjects and 100 gesture classes.
- Maintained **> 92% reconstruction fidelity** under sensor dropout and missing modalities via a reward function balancing reconstruction error ($MSE < 0.05$) and sparsity ($< 40\%$ active features per timestep). Reduced **computational overhead by 2.3×** compared to full feature baselines while achieving zero-shot adaptation to simulated sensor failures without retraining.

Project 2: Goal-Conditioned RL in Learned Embedding Space

- Achieved **< 12% reconstruction error** on unseen gesture classes through **zero-shot transfer** using a goal conditioned SAC policy agent operating in a **compressed 5D embedding space** from 24D raw features.
- Resolved sparse reward problems across 20 gesture classes via **Hindsight Experience Replay**, improving sample efficiency by 3.5× compared to standard SAC.
- Demonstrated policy generalization with **94% goal-reaching** success on trained classes and **88% on unseen** classes, while maintaining average inference latency $< 15\text{ms}$ per control step.

INDUSTRY EXPERIENCE

MathWorks | Parallel Code Generation Team

May 2023 - Aug 2023

Engineering Development Group Intern

Natick, MA

- Developed machine learning models in MATLAB and Simulink and translated these scripts into **100% production-ready** Embedded C/C++ code, demonstrating applied research techniques.
- Implemented **5 use cases** that reinforced the **advantages** of the current approach in multi-dimensional complex matrix arithmetic.
- Engineered **3 use cases** to highlight algorithmic weaknesses while showcasing the benefits of Look-ahead Superword Level Parallelism.
- Devised a general template for the **Look-ahead Superword Level Parallelism** algorithm.

MathWorks | Domain Specific Code Generation Team

May 2022 - Aug 2022

Engineering Development Group Intern

Natick, MA

- Collaborated on Embedded Coder and Domain Specific Code Generation projects, emphasizing the integration of deep learning models within system design.
- Constructed a **5-layer CNN** for brain tumor classification that outperformed a custom 4-layer CNN and VGGNet16 (**82% accuracy** versus 71% and 66% without preprocessing) using MATLAB, Simulink, adam and sgdm optimizers with Neighborhood Processing Subsystem.
- Automated the translation of MATLAB layers to Simulink subsystems via a custom script, achieving **90% code reusability**.
- Engineered a fully reusable **filter for edge detection** in image preprocessing using Simulink's Neighborhood Processing Subsystem.
- Established foundational structures for a '**convolution block**' in Simulink that abstracts **filter/convolution** parameters in deep learning models.

SELECTED PUBLICATIONS (Total 31+ publications)

- P. Olikkal, et. al., “*RL-Synergies: A RL framework for adaptive motor synergy selection across sensing modalities*”, review in BSPC
- P. Olikkal, et. al., “*RL-based adaptive synergy selection for Robust Hand Motion Reconstruction*”, under review in IEEE BSN